Habib University
Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Spring 2026
Instructor – Dr. Aamir Hasan - Section L4

Time = 20 Minutes **Quiz No 08** [20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning.

CLO No 04 - Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3)

Question 1 [14 points] [Joint Probability Density Function, 2-dimensional]

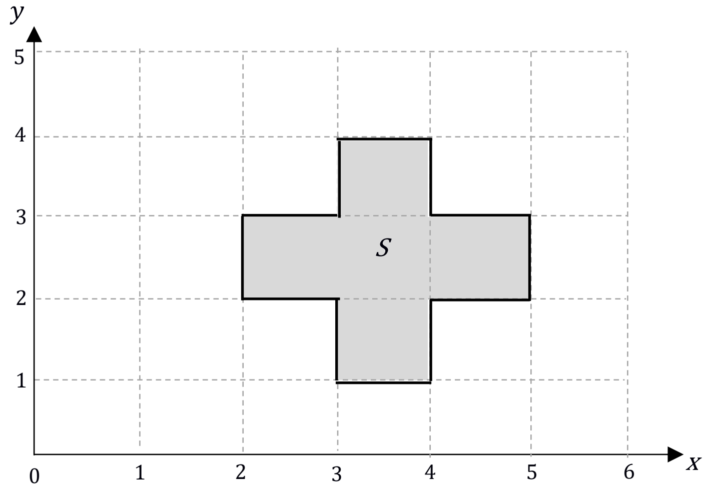
Random Variable X and **Y** are jointly continuous on the shaded region S (see below). The joint probability density function $f_{X,Y}(x,y)$ is uniformly distributed over the region S. **Note** - provide necessary justification for all the following parts.

Part a [02 points] Determine the joint probability density function $f_{X,Y}(x,y)$.

Part b [02 points] Calculate the probability of Event A, where A is described where random variable $Y = X$.

Part c [02 points] Calculate the probability of Event B, where B is given by 2.5 < $\mathbf{X}$ < 4.5 and 1 < $\mathbf{Y}$ < 4.

Part d [08 points] Calculate and plot $f_X(x)$ and $f_Y(y)$.



CLO No 04 - Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3)

Question 2 [06 points] [Joint Probability Density Function, 3-dimensional]

Random Variable X, Y and **W** are jointly continuous on a region S. The joint probability density function $f_{X,Y,W}(x, y, w)$ is uniformly distributed over the 3-dimensional cube, S, given by $0 \leq X \leq 2,\ 0 \leq Y \leq 2,\ 0 \leq W \leq 2.$

Part a [03 points] What is the interpretation of the probability density function, $f_{X,Y,W}(x, y, w)$ at a point (x, y, w) in S. *Note – make sure the interpretation explains everything discussed in the class.*

Part b [03 points] Calculate the probability of an Event A, where A is described by, $\frac{1}{2} \leq X \leq \frac{3}{2},\ 0 \leq Y \leq \frac{3}{2},\ 0 \leq W \leq 2.$